FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
TOPIC: DERIVATIVE AS A RATE MEASURE-HA

CLASS: XII **NAME :**

1. The side of a square sheet is increasing at the rate of 4 cm/min. At what rate is the area increasing when the side is 8 cm long?
2. An edge of a variable cube is increasing at the rate of 3 cm/sec. How fast is the volume of the cube increasing when the edge is 10 cm long?
3. The side of a square is increasing at the rate of 0.2 cm/sec. Find the rate of increase of the perimeter of the square.
4. The radius of a circle is increasing at the rate of 0.7 cm/sec. What is the rate of increase of its circumference?
5. The radius of a spherical soap bubble is increasing at the rate of 0.2 cm/sec. Find the rate of increase of its surface area, when the radius is 7 cm.
6. The radius of an air bubble is increasing at the rate of 0.5 cm/sec. At what rate is the volume of the bubble increasing when the radius is 1 cm?
7. The volume of a spherical balloon is increasing at the rate of 20 cm^3/sec. Find the rate of its surface area at the instant when its radius is 8 cm.
8. The radius of a cylinder is increasing at the rate of 2 m/sec and the height is decreasing at the rate of 3m/sec. Find the rate of change of volume when its radius is 3 metres and height is 5 metres.
9. The surface area of a spherical bubble is increasing at the rate of 2 cm^2/s. When the radius of the bubble is 6 cm, at what rate is the volume of the bubble is increasing?

10. The length x of a rectangle is decreasing at the rate of 5 cm/sec and the width y is increasing at the rate of 4 cm/sec. When x = 8 cm/sec and y = 6 cm, find the rate of change of

    i) the perimeter

    ii) the area of the rectangle.

11. A stone is dropped into a quite lake and waves move in circles at a speed of 4 cm/sec. At the instant when the radius of the radius of the circular wave is 10 cm, how fast is the enclosed area increasing?

12. A man 180 cm tall walks at a rate of 2 m/sec away from a source of light that is 9 m above the ground. How fast is the length of his shadow increasing when he is 1 m away from the pole?

13. A particle moves along the curve $y = x^2+2x$. At what points on the curve are the x and y coordinates of the particle changing at the same rate?

14. Water is running into an inverted cone at the rate of π cubic metres per minute. The height of the cone is 10 metres, and the radius of its base is 5 m. how fast the water level is rising when the water stands 7.5 m above the base.

15. Sand is being poured onto a conical pile at the constant rate of 50 cm^3/minute such that the height of the cone is always one half of the radius of its base. How fast the height of the pile increasing when the sand is 5 cm deep.

16. Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same rate.

FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
TOPIC: DERIVATIVE AS A RATE MEASURE-HA-2

CLASS: XII **NAME:**

1. A spherical ball is of salt is dissolving in water in such a manner that the rate of decrease of the volume at any instant is proportional to the surface. Prove that the radius is decreasing at a constant rate.

2. If the area of a circle increases at a uniform rate, then prove that perimeter varies inversely as the radius.

3. A kite is moving horizontally at a height of 151.5 m. If the speed of kite is 1 10 m/s, how fast is the string being let out, when the kite is 250 m away from the boy who is flying the kite? (The height of boy is 1.5 m.)

4. Find the approximate volume of metal in a hollow spherical shell whose internal and external radii are 3 cm and 3.0005 cm respectively.

5. A man, 2 m tall, walks at the rate of $1\frac{2}{3}$ m/s towards a street light which is $5\frac{1}{3}$ m above the ground. At what rate is the tip of his shadow moving? At what rate is the length of the shadow changing when he is $3\frac{1}{3}$ m from the base of the light?

6. A swimming pool is to be drained for cleaning. If L represents the number of litres of water in the pool t seconds after the pool has been plugged off to drain and $L = 200(10-t^2)$. How fast is the water running out at the end of 5 seconds and what is the average rate at which the water flows out during the first 5 seconds?

7. The volume of a spherical balloon is increasing at the rate of 20 cm^3/sec. Find the rate of its surface area at the instant when its radius is 8 cm.

8. The radius of a cylinder is increasing at the rate of 2 m/sec and the height is decreasing at the rate of 3 m/sec. Find the rate of change of volume when its radius is 3 metres and height is 5 metres.

9. The two equal sides of an isosceles triangle with fixed base b are decreasing at rate of 3 cm per second. How fast is the area decreasing when the two equal sides are equal to the base?

10. The total cost C(x) in rupees associated with the production of x units of an item is given by $C(x) = 0.007x^3 – 0.03x^2 + 15x + 4000$. Find the marginal cost when 17 units are produced.

11. The total revenue in rupees received from the sale of x units of a product is given by $R(x) = 13x^2 + 26x + 15$. Find the marginal revenue when x=7.

12. A balloon, which always remains spherical has a variable radius. Find the rate at which its volume is increasing with the radius when the later is 10 cm.

13. A ladder 5 m along is leaning against a wall. The bottom of the ladder is pulled along the ground, away from the wall, at the rate of 2 cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4 m away from the wall?

FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
TOPIC: APPLIED PROBLEMS IN MAXIMA AND MINIMA -1

CLASS: XII **NAME:**

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1. Divide 64 into two parts such that the sum of the cubes of two parts is minimum.
2. Divide 15 into two parts such that the square of one multiplied with the cube of the other is minimum.
3. Find two positive numbers x and y such that their sum is 35 and the product $x^2 y^5$ is a maximum.
4. Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.
5. A wire of length 25 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a circle. What should be the lengths of the two pieces so that the combined area of the square and the circle is minimum.
6. A wire of length 20 m is to be cut into two pieces. One of the pieces will be bent into the shape of a square and the other into shape of an equilateral triangle. Where the wire should be cut so that the sum of the areas of the square and triangle is minimum.?
7. A large window has the shape of a rectangle surmounted by an equilateral triangle. If the perimeter of the window is 12 m find the dimensions of the rectangle that will produce the largest area of the window.
8. Given the sum of the perimeters of a square and circle, show that the sum of there area is least when one side of the square is equal to diameter of the circle.

9. Show that the cone of the greatest volume which can be inscribed in a given sphere has an altitude equal to 2/3 of the diameter of the sphere.

10. An open tank is to be constructed with a square base and vertical sides so as to contain a given quantity of water. Show that the expenses of lining with lead will be least, if depth is made half of width.

11. A closed cylinder has volume 2156 cm^3. What will be the radius of its base so that its total surface area is minimum.

12. Two sides of a triangle have lengths a and b and the angle between them is α.
 a) What value of α will be maximize the area of the triangle?
 b) Find the maximum area of the triangle also.

13. The sum of the perimeter of a circle and square is k, where k is some constant. Prove that the sum of their areas is least when the side of square is double the radius of the circle.

14. A tank with rectangular base and rectangular sides, open at the top is to be constructed so that its depth is 2 m and volume is 8 m^3. If building of tank costs ₹ 70 per sq. metres for the base and ₹ 45 per square metre for sides. What is the cost of least expensive tank?

15. A metal box with a square base and vertical sides is to contain 1024 cm^3. The material for the top and bottom costs ₹ 5 per cm^2 and the material for the sides costs ₹ 2.50 per cm^2. Then, find the least cost of the box.

FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
TOPIC: INCREASING AND DECREASING FUNCTIONS-2
CLASS: XII **NAME:**

Find the intervals in which f(x) is i) increasing
ii) decreasing:

1. $f(x) = 2x^3 - 12x^2 + 18x + 15$.
2. $f(x) = 6 - 9x - x^2$.
3. $f(x) = 5 + 36x + 3x^2 - 2x^3$.
4. $f(x) = (x+1)^3(x-3)^3$
5. $f(x) = (x-1)(x-2)^2$
6. $f(x) = \frac{x^4}{4} + \frac{2}{3}x^3 - \frac{5}{2}x^2 - 6x + 7$
7. $f(x) = x^3 + \frac{1}{x^3}$
8. $f(x) = \log(1+x) - \frac{x}{x+1}$.
9. Show that $f(x) = \sin x - \cos x$ is an increasing function on $\left(\frac{-\pi}{4}, \frac{\pi}{4}\right)$
10. Find the intervals in which $f(x) = \sin x - \cos x$, where $(0, 2\pi)$ is increasing or decreasing.
11. Show that the $f(x) = (x-1)e^x+1$ is an increasing function for all $x > o$.
12. a) Prove that the function f given by $f(x) = \log(\cos x)$ is strictly increasing on $\left(0, \frac{\pi}{2}\right)$ and strictly decreasing on $\left(-\frac{\pi}{2}, \pi\right)$.

    b) Find the intervals in which $f(x) = (x+2)e^{-x}$ is increasing or decreasing.